

Example 2

A Diesel cycle operates on air with a low pressure of 0.1 MPa & a low temperature of 15°C. If the compression ratio is 16 & the heat supplied is 1800 kJ/kg. Compute:

1. The pressure, the temperature & the specific volume for each state.
2. The cutoff ratio.
3. The thermal efficiency.
4. The mean effective pressure.

Analysis

Process 1→2 / Isentropic compression ($s=c$)

State 1: $T_1 = 15^\circ\text{C} = 288\text{ K}$
 $P_1 = 0.1\text{ MPa} = 100\text{ kPa}$

($PV=RT$) $V_1 = \frac{RT_1}{P_1} = 0.827\text{ m}^3/\text{kg}$

State 2: Isentropic, Ideal gas situation

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = r^{\gamma-1} = 16^{\gamma-1}$$

$$T_2 = 873.5\text{ K}$$

$$\Rightarrow \frac{P_2}{P_1} \left(\frac{V_1}{V_2}\right)^\gamma = r^\gamma = 16^\gamma$$

$$P_2 = 4.85\text{ MPa}$$

$$\Rightarrow \frac{V_1}{V_2} = r \Rightarrow V_2 = \frac{V_1}{r} = 0.0517\text{ m}^3/\text{kg}$$

Process 2→3 Isobaric process ($P=c$)

State 3: $P_3 = P_2 = 4.85\text{ MPa}$

1st law of THD for process $\sum E_{in} - \sum E_{out} = \Delta E_{system}$
 $(\dot{Q}_{in} + \dot{W}_{in}) - (\dot{Q}_{out} + \dot{W}_{out}) = \Delta U + \Delta KE + \Delta PE$

$$\dot{Q}_{in} = C_p(T_3 - T_2) = 1800\text{ kJ/kg}$$

$$\Rightarrow T_3 = \frac{\dot{Q}_{in}}{C_p} + T_2 = \frac{1800}{1.0035} + 873.5\text{ K}$$

$$T_3 = 2667\text{ K}$$

$$V_3 = ? \text{ considering } \frac{P_3}{T_3} = \frac{P_2}{T_2} \Rightarrow \frac{RT_3}{V_3} = \frac{RT_2}{V_2}$$

Solution

Given

Diesel cycle

$\gamma = 1.4$

$P_1 = 0.1\text{ MPa}$

$T_1 = 15^\circ\text{C}$

$r = 16$

$\dot{Q}_{in} = 1800\text{ kJ/kg}$

Required

1. (P, T, V) for each state

2. r_c

3. $\eta_{th, Diesel}$

4. MEP

Assumptions

- The cold air standard assumptions are applicable, thus air can be assumed to have constant sp heats at room temperature.
- ΔKE & ΔPE are negligible.

$C_p =$

$C_v =$

@ 288 K $C_p = 1.003$

250 - 1.003 0.716

288 - ?

300 - 1.005 0.718

$$\frac{38}{50} = \frac{288 - 250}{300 - 250} = \frac{C_p - 1.003}{1.005 - 1.003} = \frac{C_v - 0.716}{0.718 - 0.716}$$

$$C_p = 1.00452$$

$$\frac{V_3}{V_2} = \frac{T_3}{T_2}$$

$$\Rightarrow V_3 = 0.1578\text{ m}^3/\text{kg} : V_3 > V_2$$

⇒ The piston moved down in heat addition

Process 3→4 Isentropic expansion ($s=c$)

↳ Ideal gas & Isentropic relations

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\gamma-1} = \left(\frac{0.827}{0.1578}\right)^{\gamma-1} \quad \left| \begin{array}{l} V_1 = V_4 \\ = 0.827\text{ m}^3/\text{kg} \end{array} \right.$$

⇒ $T_4 = 1375\text{ K}$ ⇒ still high; it can be recovered before rejection to produce? electric power for car: ...

$$\Rightarrow \frac{P_3}{P_4} = \left(\frac{V_4}{V_3}\right)^\gamma$$

$$\Rightarrow P_4 = 0.48\text{ MPa}$$

$$V_4 = V_1 = 0.827\text{ m}^3/\text{kg}$$

2. r_c (cutoff ratio)

$$r_c = \frac{V_3}{V_2} = \frac{0.1578}{0.0517} = 3.05$$

$$3. \eta_{th, Diesel} = \frac{W_{net}}{\dot{Q}_{in}} = 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}} = 1 - \frac{C_v(T_4 - T_1)}{C_p(T_3 - T_2)} = 1 - \frac{0.716(1375 - 288)}{1.00452(2667 - 873.5)} = 56.7\%$$

$$\text{and } \dot{Q}_{out} = C_v(T_4 - T_1) = 0.716(1375 - 288) = 778.3\text{ kJ/kg}$$

When variable specific heat is used

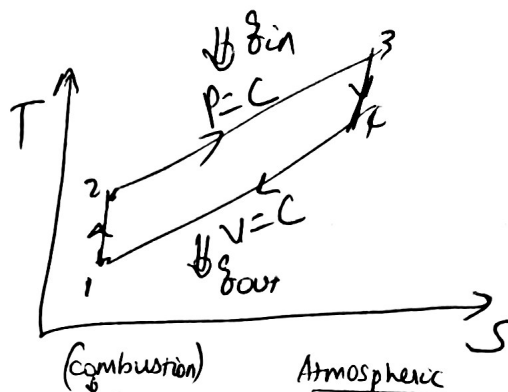
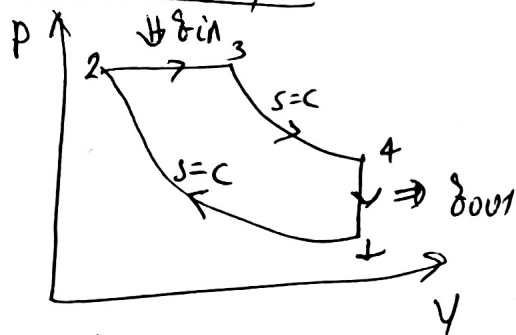
$$P_2 = \frac{P_1}{r} \Rightarrow P_2 = \frac{P_1}{16}$$

$$\left(\frac{V_2}{V_1}\right)^{\gamma} = \left(\frac{V_2}{V_1}\right)^{\gamma}$$

For same r
 $\eta_{th, Diesel} > \eta_{th, Otto}$
 For constant maximum P & same heat input

$(\eta_{th})_{Diesel} > (\eta_{th})_{Otto}$
 For same max P & T
 $(\eta_{th})_{Diesel} > (\eta_{th})_{Otto}$
 For same max P & output
 $(\eta_{th})_{Diesel} > (\eta_{th})_{Otto}$

Diesel cycle



knowing $r = \frac{V_1}{V_2}$ & $r_c = \frac{V_3}{V_2}$ & P_1, T_1

(1→2), (3→4) isentropic processes

Relative p & v
from table

$$\left(\frac{V_2}{V_1} \right)_{s=\text{constant}} = \left(\frac{V_2}{V_1} \right) = \frac{1}{r}$$

$$\left(\frac{P_2}{P_1} \right)_{s=\text{constant}} = \frac{P_2}{P_1}$$

$$P_3 = P_2$$

From ideal gas law

$$\frac{P_3 V_3}{T_3} = \frac{P_2 V_2}{T_2}$$

1st law: $\frac{2-3: p=c}{T_3}$
 q_{in} for fixed mass

$$q - w = u_3 - u_2$$

$$w_b = \int_2^3 p dv = p v \Big|_2^3; dv \neq 0$$

$$\therefore q_{in} = h_3 - h_2 \quad \text{includes boundary work when volume changes}$$

For q_{out} :

$$q_{out} = u_4 - u_1 \quad \text{when } v=c$$

$$q_{out} = u_4 - u_1$$

$$W_{net} = q_{in} - q_{out}$$

Typical SI engines have r b/n 6:1 & 12:1.

For $k = 1.4$:

6:1 ($r=6$) $\eta_{th, Otto} = 0.5116$ (51.16%)

12:1 ($r=12$) $\eta_{th, Otto} = 0.6299$ (62.99%)

Practical trick us: ~~is~~ in reality

Real SI engines have lower η_{th} than ideal ones

$\eta_{th, Otto} = 20\% - 30\%$

Otto cycle (IC, SI engine)

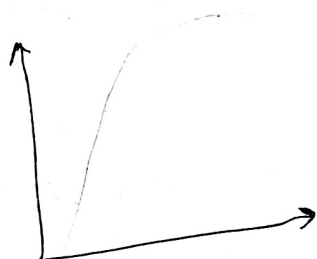
$4r - \eta_{th} \uparrow$

but r is limited by combustion knock.

↳ combustion due to high T & p before spark plug is on

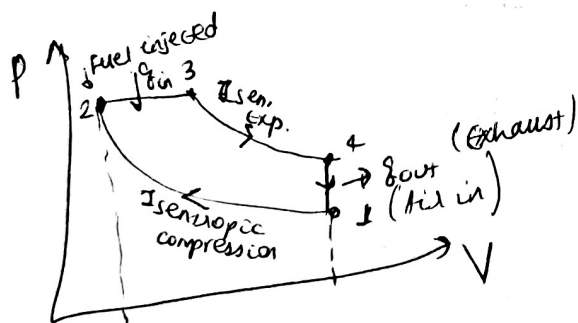
↳ Spontaneous combustion or auto ignition.

↳ result an extremely rapid pressure rise & a sharp metallic knock



Ideal Diesel or (CI Ignition process)

Constant pressure combustion



Fuel is injected after the air is compressed (at TDC), eliminating the engine knock problem.

Cut off Ratio $r_c = \frac{V_3}{V_2} = \frac{V_3}{V_2}$

$u_3 - u_2 + w_{b, out} = u_3 - u_2 + (pV_3 - pV_2)$
 $= (pV_3 - pV_2) + (u_3 - u_2) = (pV_3 + u_3) - (pV_2 + u_2)$

First law

2-3 = $q_{in} - w_{b, out} = u_3 - u_2 \Rightarrow q_{in} = (pV_3 - pV_2) + (u_3 - u_2)$
 $q_{in} = h_3 - h_2 = c_p(T_3 - T_2)$

1-4 = $-q_{out} - w = u_4 - u_1$
 $-q_{out} = u_4 - u_1 = c_v(T_4 - T_1)$

$\eta_{th, Diesel} = \frac{w_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}}$

$\eta_{th, Diesel} = 1 - \frac{1}{r^{k-1}} \left[\frac{r_c^k - 1}{k(r_c - 1)} \right]$

$\therefore \eta_{th, Diesel} \leq \eta_{th, Otto}$

$\eta_{th, Diesel} \rightarrow \eta_{th}$ much $r \rightarrow \eta_{th} \uparrow$
 30%-40% & 100% pure fuel

& according to design

Typical SI engines have r b/n 6:1 & 12:1.

For $k = 1.4$:

6:1 ($r=6$) $\eta_{th, Otto} = 0.5116$ (51.16%)

12:1 ($r=12$) $\eta_{th, Otto} = 0.6299$ (62.99%)

Practical trick us: ~~to~~ in reality

Real SI engines have lower η_{th} than ideal ones

$\eta_{th, Otto} = 20\% - 30\%$

Otto cycle (IC, SI engine)

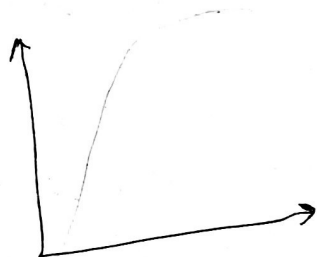
$4r - \eta_{th} \uparrow$

but r is limited by combustion knock.

↳ combustion due to high T & p before spark plug is on

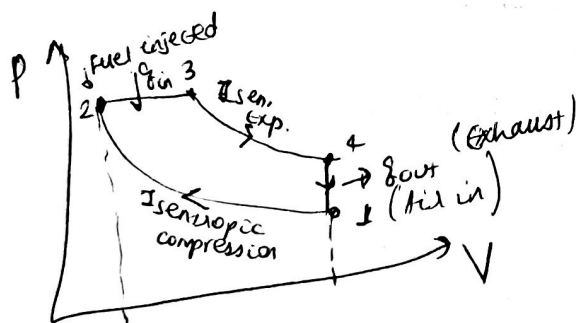
↳ Spontaneous combustion or auto ignition.

↳ result an extremely rapid pressure rise & a sharp metallic knock



Ideal Diesel or CI Ignition process

Constant pressure combustion



Fuel is injected after the air is compressed (at TDC), eliminating the engine knock problem.

Cut off Ratio $r_c = \frac{V_3}{V_2} = \frac{V_3}{V_2}$

$$u_3 - u_2 + w_{b, out} = u_3 - u_2 + (pV_3 - pV_2)$$

$$= (pV_3 - pV_2) + (u_3 - u_2) - (pV_2 - u_2) + (pV_3 - u_3)$$

$$= - (pV_2 - u_2) + (pV_3 - u_3)$$

$\eta_{th, Diesel} = \frac{w_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}}$

$\eta_{th, Diesel} = 1 - \frac{1}{r^{k-1}} \left[\frac{r_c^k - 1}{k(r_c - 1)} \right]$

$\therefore \eta_{th, Diesel} \leq \eta_{th, Otto}$

First law

2-3 = $q_{in} - w_{b, out} = u_3 - u_2 \Rightarrow q_{in} = u_3 - u_2$

$$q_{in} = h_3 - h_2 = c_p(T_3 - T_2)$$

1-4 = $-q_{out} - w_{b, out} = u_4 - u_1$

$$-q_{out} = u_4 - u_1 = c_v(T_4 - T_1)$$

$\eta_{th, Diesel}$ much $r \rightarrow \eta_{th} \uparrow$
30%-40% & less pure fuel

& according to design